covers data representation

DLP AutoEncoders

till now we studied supervised learning.
↳ Identifying dog, cat etc,
↳ Data labelling.

Concept of Encoder Decoder :-

Data → Neural Network → Encoded representation → Neural Network → Decoder

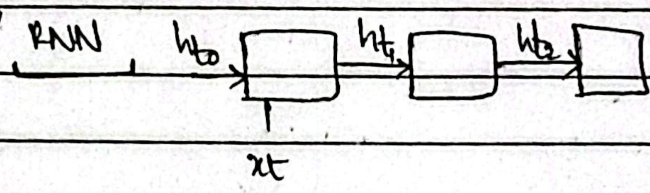
Neural Network ↓ role :- Transformation

Neural Network ↓ role :- detransformation

Applications :-

Language Translation | Sentiment Analysis

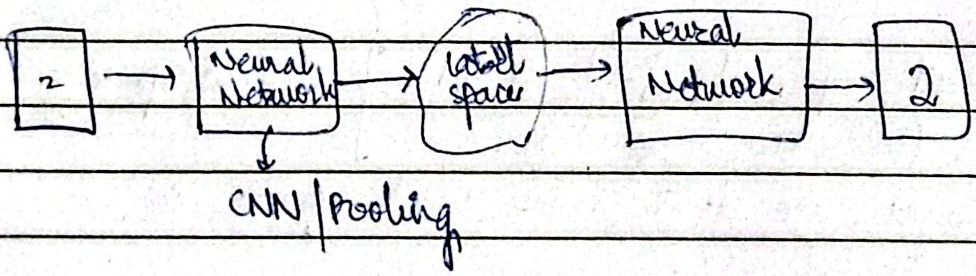
"I am fine"
↳ RNN

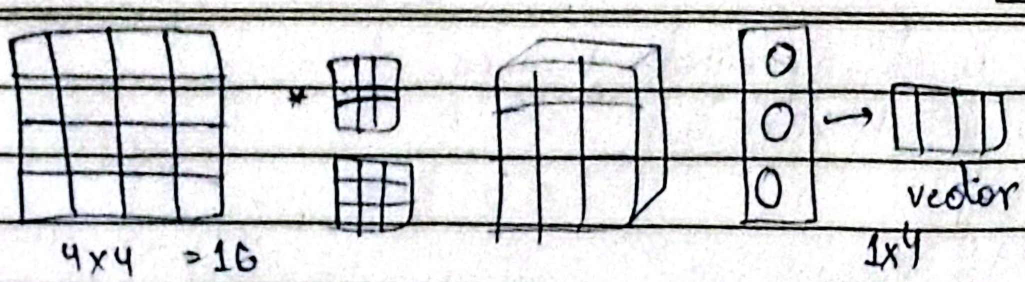


Auto Encoders :-

- Learn data patterns
- Recreate new data.

Image based Auto encoder :-





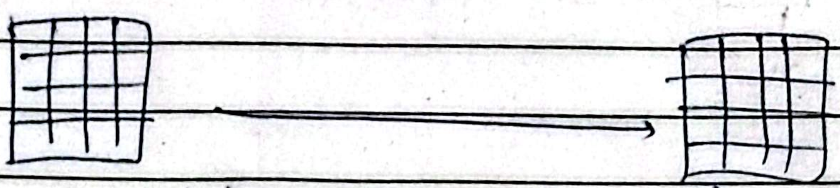
goal :- understand data & create / generate new data.

Why we encode?

16 pixels $\xrightarrow{\text{encode}}$ [| | |] vector of 4 pixels.

to prevent overfitting, if encoded vector is also 4x4.

Trade-off between larger encoded value & smaller encoded value.



X → data $\hat{X}$ → data created

Loss function :-

$$L = |x - \hat{x}|$$

Latent space :-

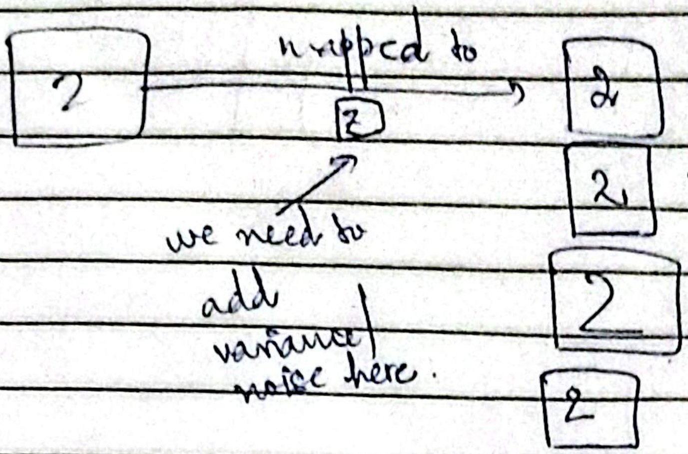
we can represent any data point as 2d (x,y) (latitude, longitude) } → latent space

:- but for images, we don't know latent space size.

Date:

→ we need variations of data created.

for ex :-



we need to add variance/noise here.

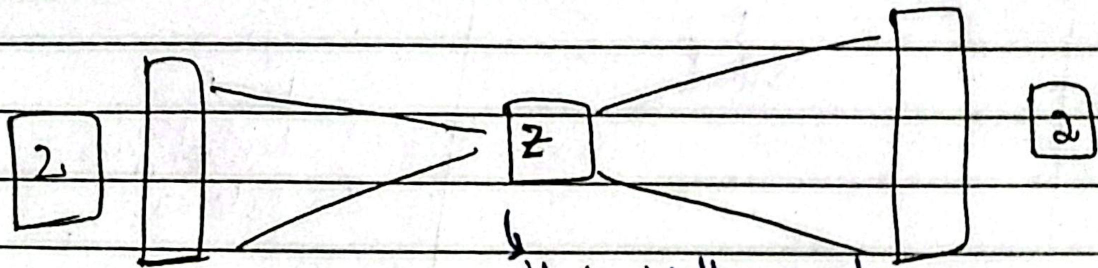
we need different variations but, auto encoder generates fixed mappings, generating same image each time.

that's why we need VAEs.

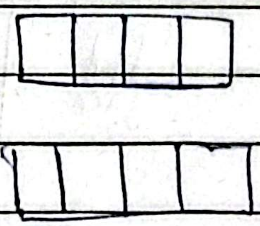
X → z → $\hat{X}$

(image) data — latent space/encoded — data created

Variational Auto Encoders :-



can't be totally random



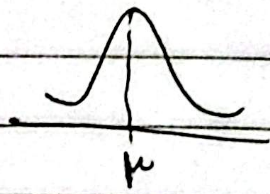
this data distribution should be bounded, not fully random.

Normal distribution

$\mu = 0$

std. dev = 1

$\sigma = 1$



this should follow Normal distribution

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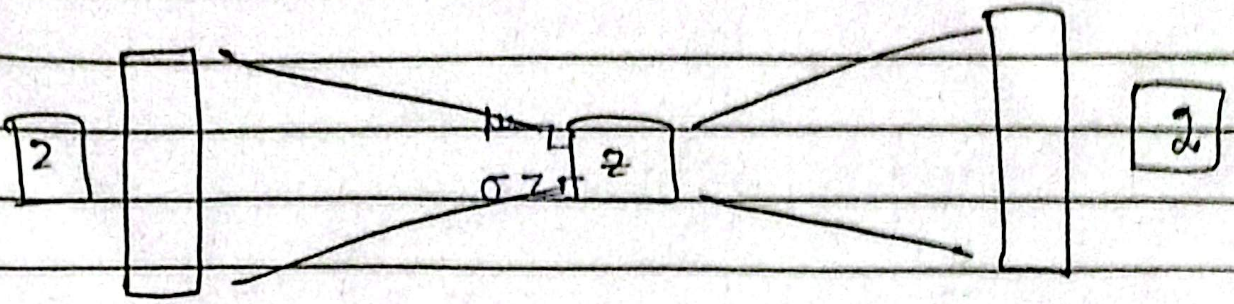
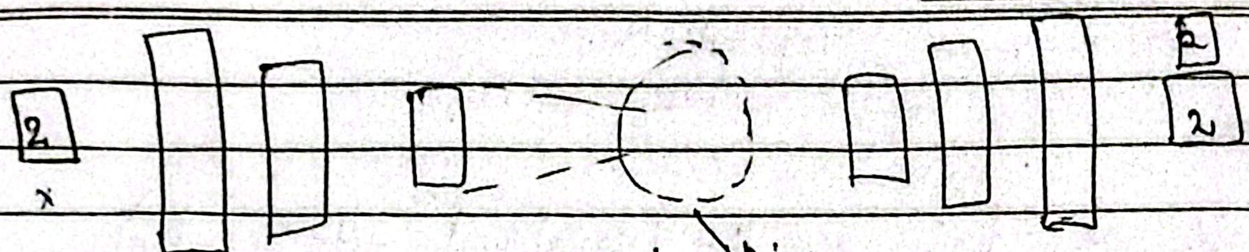


Smile

Angry face

Smile	Auto encoder	VAE (variational Auto Encoder)
Angry face	Angry face 0	Angry face interval
Neutral	Neutral	Neutral face interval
Happy face	Happy face	Happy face interval

Signature ____



prior

mean = 0
std. dev = 1

Regularization:-

Encoder

$q_\phi (z/x)$
↓
weights

1) Continuity → two points which are close in the latent space should also be close in the decoded space.
2) Completeness

Meaningful output

$L(\phi, \theta, x)$ ↓ weights at the encoder side → weights at the decoder side

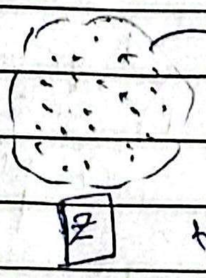
$P_\theta (\hat{x}/z)$

Loss = Reconstruction loss + Regularization loss
↓
output kitna regularized hai
→ we are applying regularization by Gaussian/Normal Distribution.

∴ We need non-deterministic, variance response in real-world example → GenAI → chatgpt → different type of response each time.

K-L Divergence ← Regularization loss

How do we train a VAE (variational Auto encoder).



This whole embedding space is a whole distribution (not any single fixed point).

toe ham kis static point pr backpropagate karenge.

→ For back propagation, we need a static (reference) point.

Technique :- Re-parametrization.

we introduce a sampling layer.

random

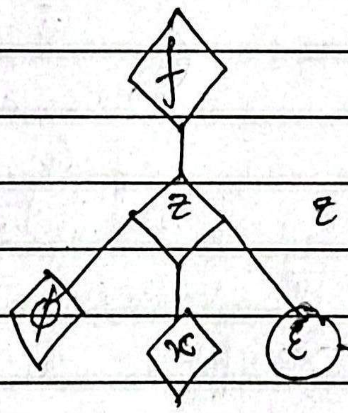
$$z = \boxed{\mu + \sigma} \odot \varepsilon \qquad \varepsilon \in N(0,1)$$

XOR

fixed point

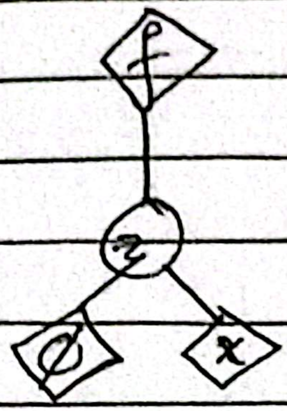
ε → between 0 & 1. It's there to introduce randomness.

◇ = Deterministic boxes



$z = g(\phi, x, \varepsilon)$

ε → Non-deterministic / stochastic



for training | backpropagation

$z = \mu + \sigma \odot \varepsilon$ ← random

↑ XOR

$$\boxed{z = g(\phi, \theta, x)}$$

ϕ: set of weights for encoder

θ: set of weights for decoder

x: inputs

K-L Divergence -

learned by VAE (Auto encoder)

↳ measures how different the learned latent distribution is from a standard normal distribution.

$p(z) = N(0,1)$

$$D_{KL}\left(q_\phi(z|x) \,\|\, p(z)\right) = -\frac{1}{2}\sum_{j=1}^{K}\left(1+\log(\sigma_j^2)-\mu_j^2-\sigma_j^2\right)$$

$q_\phi(z|x)$ → distribution learned by the encoder.

$p(z)$ → target normal distribution; mean = 0; variance = 1.

$\sum_{j=1}^{K}$ KL - divergence is corporated for each dimension of the latent vector & then summed.

μ_j :→ mean of the learned distribution.

$-\mu_j$:- penalize if mean is different from 0.
(minus)

σ_j :- variance of the learned distribution. penalize if variance is large.

$\log(\sigma_j^2)$ = prevents the variance from collapsing to small value.

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→ If KL = 0, then it's closer to standard normal distribution.

→ If KL is large, then its very far away from standard normal distribution.